

Research Notes on Sigmoid function

Sigmoid function

>> Section 1

1st kind A unitary sigmoid function of the first kind is a bounded increasing function that approaches its lower and upper asymptotes monotonically, without oscillation. This class includes many of the standard sigmoid functions used in statistics, biomathematics, and engineering, such as the logistic function and related generalizations.

>> Section 2

Many natural processes, such as those of complex systems learning curves, exhibit a progression from small beginnings that accelerates and approaches a climax over time. When a specific mathematical model is lacking, a sigmoid function is often used. The van Genuchten–Gupta model is based on an inverted S-curve and applied to the response of crop yield to soil salinity. Examples of the application of the logistic S-curve to the response of crop yield (wheat) to both the soil salinity and depth to water table in the soil are shown in modeling crop response in agriculture. In artificial neural networks, non-smooth functions are sometimes used for efficiency; these are known as hard sigmoids. In audio signal processing, sigmoid functions are used as waveshaper transfer functions to emulate the sound of analog circuitry clipping. In Digital signal processing in general, sigmoid functions, due to their higher order of continuity, have much faster asymptotic rolloff in the frequency domain than a Heaviside step function, and therefore are useful to smooth discontinuities before sampling to reduce aliasing. This is, for example, used to generate square waves in many kinds of Digital synthesizer. In biochemistry and pharmacology, the Hill and Hill–Langmuir equations are sigmoid functions. In computer graphics and real-time rendering, sigmoid functions are used to blend colors or geometry between two values, producing smooth transitions without visible seams or discontinuities. Titration curves between strong acids and strong bases have a sigmoid shape due to the logarithmic nature of the pH scale. The logistic function can be calculated efficiently by utilizing type III Unums. A hierarchy of sigmoid growth models with increasing complexity (number of parameters) was built with the primary goal to re-analyze kinetic data, the so-called N-t curves, from heterogeneous nucleation experiments, in electrochemistry. The hierarchy includes at present three models, with 1, 2, and 3 parameters, if not counting the maximal number of nuclei N_{max} , respectively—a $\tanh^2$ based model called α_{21} originally devised to describe diffusion-limited crystal growth (not aggregation!) in 2D, the Johnson–Mehl–Avrami–Kolmogorov (JMAK) model, and the Richards model. It was shown that for the concrete purpose, even the simplest model works, and thus it was

implied that the experiments revisited are an example of two-step nucleation with the first step being the growth of the metastable phase in which the nuclei of the stable phase form.

>> Section 3

Up to shifts and scaling, many sigmoids are special cases of $f(x) = \phi(\phi(x, \beta), \alpha)$, $\{\displaystyle f(x)=\varphi(\varphi(x,\beta),\alpha),\}$ where $\phi(x, \lambda) = \begin{cases} (1-\lambda x)^{1/\lambda} & \lambda \neq 0 \\ e^{-x} & \lambda = 0 \end{cases}$ $\{\displaystyle \varphi(x,\lambda)=\begin{cases}(1-\lambda x)^{1/\lambda}&\lambda\neq 0\\e^{-x}&\lambda=0\end{cases}\}$ is the inverse of the negative Box–Cox transformation, and $\alpha < 1$ $\{\displaystyle \alpha < 1\}$ and $\beta < 1$ $\{\displaystyle \beta < 1\}$ are shape parameters. Smooth transition function normalized to $(-1,1)$:

>> Section 4

A sigmoid function is any mathematical function whose graph has a characteristic S-shaped or sigmoid curve. A common example of a sigmoid function is the logistic function. Other sigmoid functions are given in the Examples section. In some fields, most notably in the context of artificial neural networks, the term "sigmoid function" is used as a synonym for "logistic function". Special cases of sigmoid functions include the Gompertz curve (used in modeling systems that saturate at large values of x) and the ogee curve (used in the spillway of some dams). Sigmoid functions have domain of all real numbers, with return (response) value commonly monotonically increasing but could be decreasing. Sigmoid functions most often show a return value (y axis) in the range 0 to 1. Another commonly used range is from -1 to 1. There is also the Heaviside step function, which instantaneously transitions between 0 and 1. A wide variety of sigmoid functions including the logistic and hyperbolic tangent functions have been used as the activation function of artificial neurons. Sigmoid curves are also common in statistics as cumulative distribution functions (which go from 0 to 1), such as the integrals of the logistic density, the normal density, and Student's t probability density functions. The logistic sigmoid function is invertible, and its inverse is the logit function.

>> Section 5

M0: Construction method for sigmoid functions not evident or intuitive M1: Inverse of singularity functions M2: Sigmoid functions of embedded positive functions M3: Rising a sigmoid function to the power M4: Exponentiating a sigmoid function M5: Symmetric sigmoid functions derived from asymmetric ones M6: Sigmoid functions of the reciprocal independent variable M7: Embedding a sigmoid function into other function M8: Sum of sigmoid functions

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M9: Multiplication of sigmoid functions M10: Integral of the product of an increasing and a decreasing function M11: Derivation from lambda (bell-shaped) functions M12: Integration of lambda (bell-shaped) function M13: Integration of the sum of lambda (bell-shaped) functions M14: Integration of the product of two lambda (bell-shaped) functions M15: Integration of the difference of two shifted sigmoid functions M16: Integration of the product of two shifted sigmoid functions M17: Convolution of sigmoid functions M18: Integration of the product of lambda and sigmoid function M19: Solutions of ordinary differential equations M20: Solutions of partial differential equation (PDE) M21: Solutions of functional differential equation (FDE) M22: Sum of a sigmoid function and some derivatives M23: Combination of sigmoid functions, its derivative and integral M24: Filtering sigmoid functions M25: Special cases of Gauss hypergeometric functions M26: Feedback closed-loop systems M27: Recursive functions M28: Recursive time-delayed feed-forward loops M29: Solutions of stochastic differential equation M30: Chaotic sigmoid functions Consult reference for more details.

>> Section 6

Further reading Mitchell, Tom M. (1997). Machine Learning. WCB McGraw-Hill. ISBN 978-0-07-042807-2.. (NB. In particular see "Chapter 4: Artificial Neural Networks" (in particular pp. 96–97) where Mitchell uses the word "logistic function" and the "sigmoid function" synonymously – this function he also calls the "squashing function" – and the sigmoid (aka logistic) function is used to compress the outputs of the "neurons" in multi-layer neural nets.) Humphrys, Mark. "Continuous output, the sigmoid function". Archived from the original on 2022-07-14. Retrieved 2022-07-14. (NB. Properties of the sigmoid, including how it can shift along axes and how its domain may be transformed.)